• Apply literature review and identify a research gap to formulate a problem.	**2. Literature Review (6 hrs)** 2.1. Purpose of Literature Review 2.2. Literature Resources and Techniques 2.3. Conducting a literature review 2.4. Referencing and Citation 2.5. Evaluating literature review 2.6. Identifying problem: Research question / problem statements and hypothesis 2.7. Case Study: Prepare a review paper.

1. Compare and contrast different literature sources. Evaluate the effectiveness of systematic vs. narrative review methods in IT research.

In IT research, selecting and evaluating literature involves a fundamental distinction between source types and the methodologies used to synthesize them. Literature is typically categorized as **primary** (original reports of research) or **secondary** (assembled reviews of primary sources). Choosing the right review method—**systematic** or **narrative**—is critical for establishing the credibility and scope of a research project.

Comparison of Literature Sources

Different sources offer varying levels of detail, authority, and currency:

- **Journals vs. Conferences:** In IT, **conference proceedings** often report the latest research developments because they are published more quickly than journals. However, **peer-reviewed journals** are generally considered more authoritative and undergo a more rigorous review and editing process.
- **Primary vs. Secondary Sources: Primary sources** are original observations and firsthand information, while **secondary sources** include assembled reviews, textbooks, and encyclopaedias. Secondary data is highly efficient for saving time but may not always align perfectly with specific research questions.
- **Grey Literature and Multimedia:** This includes **theses, technical reports, manuals, and online content**. Technical reports are particularly useful for communicating detailed results to specialized stakeholders in engineering and industry. Researchers are cautioned to be careful with internet-only sources, evaluating them based on the author's expertise and the publisher's reputation.

Evaluation of Review Methods in IT Research

Review methods vary significantly in their rigour, repeatability, and depth of analysis.

1. Narrative (General) Review Methods

Narrative reviews, often referred to as general surveys, provide a high-level overview of a topic.

- **Effectiveness:** They are useful for broad syntheses or when a topic is first being examined.
- **Limitations:** In IT research, narrative reviews are frequently criticized for having **limited coverage** and a lack of clear categorization. They often review a small number of studies (sometimes as few as 5 to 11) and may fail to provide detailed comparisons between different methods or algorithms.

2. Systematic Literature Review (SLR) Methods

A **Systematic Literature Review** is a rule-based research process that follows a structured sequence of steps.

- **Effectiveness:** SLRs are highly valued in IT for their **replicability** and scientific credibility. They use strict **inclusion and exclusion criteria** to filter hundreds of initial publications down to a relevant final set, ensuring quality and transparency. For instance, a systematic review on phishing detection might start with 537 papers and narrow them down to 80 high-quality studies.
- **Strengths:** SLRs allow for a **comprehensive view** by exploring multiple electronic databases (e.g., IEEE Xplore, ACM Digital Library) and extracting data into structured tables for comparative analysis. This allows researchers to identify the most effective algorithms or datasets within a specific field.
- **Limitations:** The primary drawback is the **significant effort and time** required for the multi-level screening and data extraction process.

Comparative Summary

Feature	Narrative Review	Systematic Literature Review (SLR)
Methodology	General overview, subjective selection.	Rule-based, structured, and transparent.
Scope	Often high-level with limited study counts.	Comprehensive, utilizing many databases and criteria.
Repeatability	Low; depends on the author's discretion.	High; the search and exclusion process is documented.

Best Use Case	Framing a problem or providing a broad backdrop.	Evaluating effectiveness and finding specific research gaps.

In conclusion, while narrative reviews are effective for establishing the context of a research problem, **systematic reviews are superior in IT research** for providing evidence-based evaluations of technology, identifying specific deficiencies in past literature, and establishing a rigorous foundation for new knowledge.

2. How can conducting a literature review help in identifying a research gap? Choose a specific domain and illustrate the process of finding research gap with a relevant example.

Conducting a literature review is a foundational research skill that helps determine whether a topic is worth studying and provides a framework for positioning a new study within the existing body of knowledge. It serves to assemble evidence that a chosen topic is worthwhile and ensures the researcher is not merely repeating the work of others.

How Literature Reviews Identify Research Gaps

A literature review identifies a research gap—a specific area where information or knowledge is missing—by following a systematic process often referred to as a **"deficiency model"**. This process helps researchers pinpoint exactly where past studies have fallen short and how their proposed work will fill that void. Literature reviews help find gaps in several ways:

- **Identifying Unexplored Populations:** Gaps often exist because a topic has not been explored within a particular group, sample, or geographic region.
- **Highlighting Methodological Flaws:** A review might reveal that previous studies used inadequate definitions or measures, suggesting a need for more rigorous methodology.
- **Locating Future Research Suggestions:** Researchers often conclude their papers with a "suggestions for future research" section, which explicitly points out remaining unknowns.
- **Spotting Conflicting Results:** When different studies on the same topic yield contradictory findings, a gap exists for a study that can resolve these contradictions.
- **Recognizing Outdated Information:** In fast-evolving fields, older literature may no longer apply to current technologies or societal contexts, creating a gap for updated research.

The Process of Finding a Research Gap: An Illustration

Researchers often use a **funnel approach** to narrow down a broad area of interest into a specific, "crisp" gap. This process involves reading general literature reviews, identifying "future

directions" mentioned by other authors, and double-checking findings through a full literature search.

Example Domain: IT Security (Phishing Website Detection)

In the domain of **phishing website detection**, a researcher might follow this process to find a research gap:

1. **Reviewing Broad Literature:** The researcher starts by looking at established methods like **list-based** and **similarity-based** techniques. They find these are effective but reactive, often failing to detect "zero-day" attacks (new threats).
2. **Evaluating Recent Advancements:** Moving down the "funnel," the researcher examines **Machine Learning (ML)** and **Deep Learning (DL)** studies. While these are popular, the review reveals that they often require large, high-quality datasets that are difficult to maintain.
3. **Identifying Specific Deficiencies:** By critically analyzing recent surveys, the researcher notices specific **expressions indicating gaps** in the text, such as "...has not been studied," or "...is required". For instance, a review might note that current datasets are **biased toward English-speaking users** or common platforms, leaving other languages and localized platforms underrepresented.
4. **Formulating a Unique Gap:** The researcher concludes that while ML is well-documented, there is a **notable gap** in using emerging mechanisms like **Large Language Models (LLMs)** or **graph-based models** to analyze structural relationships within web pages.
5. **Setting the Research Objective:** Based on this identified gap, the researcher proposes a study specifically focused on LLM-based detection to handle context-aware phishing attempts that traditional ML models might miss.

By the end of this process, the researcher has transformed a broad interest in "phishing" into a specific research problem that contributes new knowledge to the field by addressing an underexplored area.

3. Why is literature review important in research? How do you conduct a literature review? Explain with examples.

A **literature review** is a fundamental component of research that involves locating and summarising existing studies on a specific topic. It serves as a bridge between your proposed study and the larger academic conversation, ensuring your work contributes meaningful new knowledge.

Importance of a Literature Review

Conducting a review is critical for several reasons:

- **Justifying the Study:** It establishes whether a topic is **worth studying** and helps define the scope of the inquiry.
- **Identifying Research Gaps:** It allows researchers to pinpoint **deficiencies** in existing literature, such as unexplored populations, methodological flaws, or conflicting results, which the new study can then address.
- **Avoiding Replication:** It ensures that you do not unintentionally repeat work that has already been completed by others.
- **Providing a Framework:** It offers a benchmark for comparing your results with previous findings and provides a theoretical framework for your research questions.
- **Methodological Insight:** It reveals which **research methods** and instruments have been successful or problematic in previous studies.

How to Conduct a Literature Review

A systematic approach is recommended to capture, evaluate, and synthesise the literature effectively:

1. **Identify Keywords:** Begin with keywords found in your preliminary reading or your research topic to use in library and database searches.
2. **Search Databases:** Use online databases such as **Google Scholar, IEEE Xplore, ProQuest, or ERIC** to locate journal articles and books.
3. **Prioritise Sources:** Focus initially on **refereed journal articles** and research monographs, as these are typically the most current and authoritative.
4. **Skim and Select:** Skim through your initial findings (ideally around 50 reports) and select those that contribute significantly to your understanding.
5. **Design a Literature Map:** Create a **visual picture** (a map) of how the studies you have found are grouped. This helps you position your own study within the larger body of research.
6. **Draft Abstracts:** Write brief summaries of the most relevant articles, including the research problem, central purpose, sample/population, and key results.
7. **Organise and Write:** Assemble the review by structuring it **thematically** or around **major variables**.

Examples of Literature Review Components

- **Abstract Example:** A summary might describe a study that tested a specific model (e.g., the Biglan model) using a sample of 80 participants to find that different academic areas have different professional development needs.
- **Literature Map Example:** A map for a study on "Procedural Justice" might organise literature into categories like "Justice Perception Formation" and "Justice Effects," eventually leading to a "Need to Study" box at the bottom that identifies a specific gap, such as "Procedural Justice and Culture."
- **Identifying Gaps Example:** In the domain of **IT security**, a researcher might note that while many surveys exist on phishing detection, they often lack comprehensive coverage

of **emerging techniques** like LLM-based or graph-based mechanisms, thus motivating a new study to fill these specific gaps.

4. Explain the difference between referencing and citation. Why is proper citation essential in academic research? Give examples.

In academic research, citing and referencing are distinct but interdependent processes used to acknowledge the work of others and establish the credibility of your own study.

Difference Between Citation and Referencing

The primary difference lies in their location and level of detail:

- **Citing (In-text):** This is the act of notating when you quote, paraphrase, or use an idea from another person within the main body of your text. A citation serves as a brief "signpost" that points the reader to the original source. In numeric styles like IEEE, a citation is represented by a number in square brackets (e.g.,), whereas in author-date styles like APA, it includes the author's surname and year (e.g., Nickerson, 1999).
- **Referencing (End-of-text):** This involves creating a comprehensive list of all the sources you have cited throughout your work, usually placed at the very end. Each entry in this list provides full bibliographic details (authors, title, publisher, date) so that a reader can identify and locate the specific material you used.
- **Difference from Bibliography:** While a reference list only includes sources you directly used and cited in your text, a **bibliography** is broader, including everything you read during your research regardless of whether you cited it.

Why Proper Citation is Essential

Proper citation is a core skill for any researcher and is essential for several reasons:

- **To Avoid Plagiarism:** Plagiarism is the unethical practice of presenting someone else's words or ideas as your own. Proper citation ensures that original authors receive "appropriate credit" for their intellectual property.
- **To Establish Credibility and Integrity:** Citations demonstrate that you have read widely at an appropriate academic level and used credible, good-quality sources to support your arguments. This establishes **trust and accuracy** in your findings.
- **To Allow Differentiation:** It allows markers and readers to clearly distinguish between your original thoughts and the work of others.
- **To Enable Verification:** Providing full references allows other researchers to verify your claims, check your data, and further explore the topic, which is vital for the advancement of knowledge.

- **Legal and Professional Requirements:** Failure to cite properly can lead to serious consequences, including failing grades, loss of funding, job loss, or legal issues related to copyright infringement.

Examples of Citations and References

1. APA Style (Author-Date)

- **In-text Citation:** "Treating people as though they have knowledge that they do not have can result in miscommunication (Nickerson, 1999)."
- **Reference List Entry:** Nickerson, R. S. (1999). How we know - and sometimes misjudge - what others know: Imputing one's own knowledge to others. *Psychological Bulletin, 125*(6), 737-759.

2. IEEE Style (Numeric)

- **In-text Citation:** "The theory was first put forward in 1987."
- **Reference List Entry (Book):** I. A. Glover and P. M. Grant, *Digital Communications*, 3rd ed. Harlow: Prentice Hall, 2009.
- **Reference List Entry (Journal):** F. Yan et al., "Study on the interaction mechanism between laser and rock during perforation," *Optics and Laser Technology*, vol. 54, pp. 303-308, Dec 2013.

5. Develop two research titles, objectives, research questions, a problem statement and hypothesis on the topic **"Role of AI in medical sciences."** Also explain how a research gap can be found in this particular topic.

Based on the principles of academic writing and research design outlined in the sources, here are two sets of research components for the topic **"Role of AI in medical sciences,"** followed by an explanation of how to identify a research gap.

Research Option 1: Clinical Accuracy (Quantitative Focus)

- **Title:** AI Accuracy in Early Detection of Diabetic Retinopathy: A Comparative Study.
- **Problem Statement:** Diabetic retinopathy is a leading cause of vision loss, yet traditional manual screening by specialists is often hindered by human fatigue and limited access to experts in rural areas. While AI shows promise, there is a lack of rigorous comparative data regarding its performance against gold-standard clinical assessments in diverse population samples.
- **Objectives:** To evaluate the diagnostic sensitivity of a deep learning algorithm compared to human ophthalmologists in identifying early-stage diabetic retinopathy from retinal images.

- **Research Question:** To what extent does the implementation of an AI-based diagnostic tool improve the accuracy of early-stage diabetic retinopathy detection compared to traditional manual clinical review?.
- **Hypothesis:**
    - **Null (\(H_0\)):** There is no significant difference in the sensitivity rates between AI-based diagnostic tools and manual clinical reviews for detecting early-stage diabetic retinopathy.
    - **Alternative (\(H_1\)):** AI-based diagnostic tools achieve significantly higher sensitivity and precision rates than manual clinical reviews in the early detection of diabetic retinopathy.

Research Option 2: Ethics and Trust (Qualitative Focus)

- **Title:** Patient Perceptions of Data Privacy in AI-Driven Personalized Medical Consultations.
- **Problem Statement:** Despite the potential for AI to personalize treatment, the use of large-scale sensitive health data for training models raises critical concerns regarding patient privacy and the potential for algorithmic bias. If patients do not trust the "black box" nature of AI, the adoption of these life-saving technologies will remain limited.
- **Objectives:** To explore and understand the specific privacy concerns and trust levels that patients experience when their medical data is processed by AI-driven diagnostic systems.
- **Research Question:** How do patients describe their trust in medical recommendations generated by AI when informed about the data collection and algorithmic decision-making processes involved?.
- **Hypothesis (Predictive component):** Patients provided with Explainable AI (XAI) frameworks will report significantly higher levels of trust in medical recommendations than those receiving "black-box" AI results.

How to Find a Research Gap in AI in Medical Sciences

Identifying a research gap involves a systematic process of determining where existing knowledge ends and where your study can contribute something new. In the field of AI and medicine, you can follow these steps:

6. **Employ the "Funnel Approach":** Start by reading broad **systematic literature reviews** or "state-of-the-art" surveys on AI in medicine. Narrow your focus from "AI in Healthcare" to a specific sub-domain, such as "AI in Oncology" or "AI Ethics".
7. **Look for "Gap Indicators" in Text:** Critically analyze existing publications for specific phrases that signal deficiencies, such as:
    a. "...has not been studied/reported...".
    b. "...the key question remains...".
    c. "...further research is required to address...".

8. **Identify Specific Deficiencies:** Common gaps in medical AI research often include:
    a. **Unexplored Populations:** A model might work well for one demographic but hasn't been tested on underrepresented groups.
    b. **Methodological Flaws:** Previous studies may have used biased training data or lacked "Explainable AI" (XAI) features, leading to a "black box" problem.
    c. **Outdated Information:** In a fast-moving field, studies from only 2–3 years ago might use obsolete algorithms compared to the latest LLM-based or graph-based models.
9. **Review "Future Directions":** Most high-quality research papers conclude with a section on "Future Work" or "Future Research Scopes". These are explicit pointers provided by experts regarding what needs to be investigated next.
10. **Consult with Experts:** Discussing your preliminary findings with a supervisor or industry professional can help refine a broad interest into a "crisp" and manageable research gap.

11. What is a problem statement in research?

A **problem statement** in research (often referred to as the **research problem**) is a concise description of an issue or concern that needs to be addressed through a study. It serves as the primary justification for the project, answering the fundamental question: **"Why do this project?"**.

Core Purpose and Function

The problem statement establishes the framework for the entire study and helps readers understand how the proposed research relates to previous work. Its key functions include:

- **Defining the "Why":** It emphasizes the importance, frequency, and severity of the issue.
- **Identifying Gaps:** It explains the current limits of knowledge and where past literature has fallen short.
- **Justifying the Study:** It provides evidence that the topic is worthwhile and that the researcher is not merely repeating existing work.

Key Components of an Effective Problem Statement

A well-crafted problem statement should quickly and clearly:

- **Summarize the issue** from the vantage point of the reader.
- **Demonstrate familiarity** with prior research or work on the problem.
- **Explain the contribution** the new study will make to filling existing knowledge gaps or extending the "knowledge frontier".
- **Describe the broader impact** or benefits of the research to society, policy, or practice.

Problem Statements by Research Design

The nature of the problem being addressed varies depending on the research approach:

- **Qualitative Research:** The problem is typically **exploratory**. It is used when the researcher seeks to understand a concept or phenomenon where the variables and theory base are largely unknown.
- **Quantitative Research:** The problem is best addressed by understanding what **factors or variables** influence a specific outcome.
- **Mixed Methods Research:** The problem often requires both understanding variable relationships and exploring the topic in further depth.

The "Deficiencies Model"

In social science introductions, the problem statement is often the first part of a **"deficiencies model"**. This model consists of five steps:

1. **The research problem:** Identifying the issue.
2. **Studies addressing the problem:** Briefly reviewing existing literature.
3. **Deficiencies in those studies:** Pinpointing what is missing or flawed in past work.
4. **Significance for audiences:** Explaining who will profit from the study.
5. **The purpose statement:** Ending with the specific intent of the proposed study.

Important Distinctions

Researchers must distinguish between the **research problem**, the **purpose statement**, and the **research questions**:

- The **Research Problem** is the broad issue leading to the need for a study.
- The **Purpose Statement** is the most important statement in the proposal, establishing the study's central intent or objective.
- **Research Questions** are the specific interrogative statements that narrow the purpose and that the data collection will attempt to answer.

12. Why is a literature review important in research? As an early-stage researcher, how do you conduct a literature review? Explain.

A **literature review** is a core research skill and a fundamental component of the research process that involves locating, evaluating, and summarising existing studies on a specific topic,. For an early-stage researcher, it serves as a bridge between a personal interest and a formal academic contribution.

Importance of a Literature Review

Conducting a review is critical because it ensures that your study is grounded in the existing body of knowledge. Its primary purposes include:

- **Justifying the Study:** It shares the results of related studies and provides a framework for establishing why your research is important. It assembles evidence to support your claim that the chosen topic is **worthwhile** and will create **new knowledge**,.
- **Identifying Research Gaps:** By critically evaluating past literature, you can pinpoint **deficiencies**, such as unexplored populations, methodological flaws, or conflicting results, which your study can then address,.
- **Avoiding Unintentional Replication:** It ensures that you do not repeat work already completed by others, or provides a deliberate reason if you choose to replicate a study in a new context,.
- **Methodological Insight:** It allows you to learn which **research methods and instruments** have been successful in the past and which have proven less useful.
- **Providing a Benchmark:** The review offers a "road sign" for your study and a benchmark against which you can compare your eventual findings with previous research,.

How to Conduct a Literature Review

Early-stage researchers should follow a systematic, step-by-step approach to capture and synthesise relevant literature effectively:

1. **Identify Keywords:** Begin by identifying keywords from your research topic or preliminary readings to use in library and database searches.
2. **Search Computerised Databases:** Use major online repositories to locate material. Recommended databases include **Google Scholar**, **IEEE Xplore**, **ProQuest**, **PubMed**, **ACM Digital Library**, and **Elsevier**,,-.
3. **Prioritise Sources:** Establish a priority for the types of material you review to ensure quality and currency:
    - **1st Priority:** Broad syntheses and overviews found in encyclopaedias or journal review articles.
    - **2nd Priority: Refereed journal articles** in respected journals, starting with the most recent and working backward.
    - **3rd Priority:** Research monographs and scholarly books on the topic.
    - **4th Priority:** Recent conference papers, which often report the latest developments.
    - **5th Priority:** Highly selective doctoral dissertations.
4. **Skim and Select:** Initially locate about **50 reports** and skim them to determine if they will make a useful contribution to your understanding.
5. **Design a Literature Map:** Create a **visual picture** (a map) of how the studies you have found are grouped. This map illustrates how your particular study will contribute to the literature and positions it within the larger body of research,.
6. **Draft Abstracts:** For the most relevant articles, write brief summaries including the research problem, central purpose, sample/population, and key results,.

7. **Organise and Write:** Assemble the review by structuring it **thematically** or around **major variables** rather than just listing individual studies chronologically,. Finish by highlighting how your study will fill the identified gaps in the literature.

Key Tips for Success

- **Start Early:** Writing the literature review while the material is fresh in your mind helps clarify your thoughts and discover your true perspective.
- **Be Critical:** Do not just summarise; **critically evaluate** the existing research to identify where it falls short,.
- **Stay Focused:** Only review work that is directly related to your research questions and topic.
- **Use Proper Attribution:** Always provide a citation for every source mentioned to avoid **plagiarism** and establish your credibility as a researcher,.

13. Research question and problem_statement

In academic research, the **problem statement** and **research questions** are distinct but closely related "signposts" that guide the reader through a study's logic. While the problem statement establishes the broad justification for the work, the research questions provide the specific focus for data collection.

1. The Research Problem (Problem Statement)

A **research problem** is the specific issue, concern, or controversy that leads to the need for a study. It serves as the primary justification for the project by answering the fundamental question: **"Why do this project?"**.

- **Function:** It establishes a framework for the research, piques reader interest via a "**narrative hook**," and places the study within the context of existing scholarly dialogue.
- **Key Components:** An effective problem statement should summarize the issue from the reader’s vantage point, document its **frequency and severity**, and demonstrate the researcher's familiarity with prior work.
- **The Deficiencies Model:** In many social science introductions, the problem statement is part of a five-step "deficiencies model". This involves identifying the problem, reviewing studies that addressed it, pointing out **deficiencies** (gaps) in those studies, explaining the significance for specific audiences, and concluding with a purpose statement.
- **Knowledge Frontier:** The statement should explain current limits in knowledge and how the study will fill those gaps to extend the "**knowledge frontier**".

2. Research Questions

Research questions are interrogative statements that narrow the broad purpose of the study into specific inquiries that data collection will attempt to answer. They act as a major signpost, focusing the study's direction after the purpose statement has been established.

- **Qualitative Research Questions:** These typically include one or two **central questions**—broad inquiries asking for an exploration of the central phenomenon—followed by five to seven subquestions that narrow the focus without limiting the inquiry.
- **Quantitative Research Questions:** These inquire about the **relationships among variables** that the investigator seeks to understand. They often focus on comparing groups or relating independent and dependent variables.
- **Mixed Methods Research Questions:** These are "**hybrid**" or "**integrated**" questions that directly address the mixing or integration of the quantitative and qualitative strands of the study.

Summary of Distinctions

It is vital for researchers to distinguish between these components to ensure clarity in their proposals:

- **Research Problem:** The broad issue or "need" leading to the study.
- **Purpose Statement:** The overall intent or "objective" of the study.
- **Research Questions:** The specific "what" or "how" that the researcher will measure or explore to answer the problem.